

MALAYSIA BID ROUND 2026 (MBR 2026)

BID PROPOSAL — EXPLORATION BLOCKS & DISCOVERED RESOURCE OPPORTUNITIES

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1. EXECUTIVE SUMMARY

PETRONAS, through Malaysia Petroleum Management (MPM), launched the Malaysia Bid Round 2026 (MBR 2026) on 10 February 2026 under the theme “**Advantaged Energy: Accelerating and Shaping Tomorrow.**” The round offers **15 opportunities** — 9 exploration blocks and 6 Discovered Resource Opportunities (DROs) — across three major basin provinces: the mature **Malay Basin**, the emerging **West Sarawak Basin**, and the frontier **Sandakan Basin**.

Key Parameters

Parameter	Value
Total Opportunities	15 (9 exploration + 6 DRO)
Basins	Malay, West Sarawak, Sandakan
Fiscal Terms	Enhanced Profitability Terms (EPT) PSC for exploration; Small Field Asset (SFA) PSC for DROs
Bid Deadline	1 October 2026 (all except SB304: 30 June 2027)
PSC Award Target	February 2027
PSC Signing Target	April 2027
Prerequisite	PETRONAS myPROdata subscription
Data Rooms	Open since 10 February 2026

Recommendation Summary

RECOMMENDED TO BID: - **PM447** — Malay Basin, near-field with new play ideas (Pre-Tertiary carbonates) - **PM448** — Malay Basin, near-field, shallow Pre-Triassic targets - **Cempaka Cluster (DRO)** — Small field, ready-to-develop, low-risk

CONSIDER WITH PARTNERS: - **SK404** — West Sarawak, high potential but requires significant capital - **SK331** — West Sarawak, emerging play with 3D seismic data - **Enggang Cluster (DRO)** — Proven fields, requires redevelopment expertise

NOT RECOMMENDED: - **SB304** — Sandakan Basin frontier, extreme geological risk, no 3D seismic - **PM519/PM520** — Deep Pre-Tertiary targets, very high depth to basement risk

2. MBR 2026 PORTFOLIO OVERVIEW

2.1 Exploration Blocks

Block	Basin	Area (km ²)	Water Depth (m)	PSC Terms	Minimum Work Commitment
PM440	Malay	4,840	50–60	EPT	Phase 1: 3 yrs, drill 1 well
PM447	Malay	4,325	20–60	EPT	Phase 1: 3 yrs, drill 1 well + license MC3D Repro
PM448	Malay	—	—	EPT	Phase 1: 3 yrs, drill 1 well
PM519	Malay	5,629	5–75	EPT	Phase 1: 4 yrs, drill 1 well + 2D OBN
PM520	Malay	6,110	5–75	EPT	Phase 1: 4 yrs, drill 1 well + 3D TZ seismic
SK330	W. Sarawak	5,038	120–300	EPT	Phase 1: 3 yrs, drill 1 well + license Beicip MC PH2
SK331	W. Sarawak	2,835	70–200	EPT	Phase 1: 3 yrs, drill 1 well + license Beicip MC PH2

Block	Basin	Area (km ²)	Water Depth (m)	PSC Terms	Minimum Work Commitment
SK404	W. Sarawak	4,829	40–100	EPT	Phase 1: 3 yrs, drill 1 well + license Beicip MC PH1
SB304	Sandakan	7,030	0–20	EPT	Phase 1: 3 yrs, acquire 500 km ² 3D OBN + 1000 km 2D

2.2 Discovered Resource Opportunities (DRO)

DRO	Location	Fields	Fiscal Term
Cempaka Cluster	Pen. Malaysia	Saujana, Serok, Semangkok NAG, Semangkok Timur	SFA PSC
Teratai Cluster	Pen. Malaysia	Sotong, Malong, Anding, Aji, Aji Sentang, Parang, Naga Kecil, Naga Dalam	SFA PSC
Enggang Cluster	Sarawak	Salam, Benum, Patawali, Gagau	SFA PSC
NURI Funnel	Sarawak	Nuang, Rompin, KT1, Endau, Mahkota E2, Serunai	SFA PSC
Nilam Cluster	Sabah	Trusmadi, Madalon, Papar, Padas, Grafit	SFA PSC
Zamrud Funnel	Sabah	Tembungo, SW Emerald, Danum, Rusa Timur, Sipadan	SFA PSC

3. FISCAL TERMS ANALYSIS

3.1 Enhanced Profitability Terms (EPT) — Exploration Blocks

The EPT PSC represents PETRONAS's modern fiscal framework for shallow water exploration, introduced in 2021 to replace the legacy Revenue over Cost (R/C) structure.

Component	Terms
Cash Payment	10% of gross production to Federal/State Governments
Cost Recovery Ceiling	Fixed 70% of gross production
Profit Sharing	Based on Profitability Index (PI = cumulative revenue ÷ cumulative cost)
PI 1.5	Contractor receives 90% of profit oil
1.5 < PI 2.5	Contractor share interpolated 90% → 30%
PI > 2.5	Contractor receives 30% of profit oil

Assessment: The EPT terms are highly competitive globally. The fixed 70% cost recovery ceiling provides certainty, and the PI-based profit sharing rewards early exploration success. At low PI (early investment recovery), the contractor retains 90% — this is excellent for frontier exploration.

3.2 Small Field Asset (SFA) PSC — DROs

Component	Terms
Cash Payment	10% of gross production
PETRONAS Share (%)	Minimum set, biddable item, fixed throughout contract
Contractor Share (%)	% of gross production includes ALL costs (Capex, Opex, Abex) + profit margin
Equation	% + % = 90% (10% government cash payment)

Assessment: The SFA terms are deliberately simplified to attract smaller players. The biddable % means competitive bidding will drive PETRONAS’s share. The all-inclusive % structure reduces fiscal complexity — ideal for small field development where cost control is paramount.

3.3 Comparative Analysis

Fiscal Feature	Legacy R/C PSC	EPT PSC	SFA PSC
Cash Payment	10%	10%	10%
Cost Recovery	Variable (30–70%)	Fixed 70%	Simplified
Profit Sharing	R/C ratio dependent	PI-based	Biddable /
Complexity	High	Medium	Low
Best For	Large fields	Exploration	Small fields

4. EXPLORATION BLOCKS — GEOLOGICAL ASSESSMENT

4.1 MALAY BASIN BLOCKS (PM440, PM447, PM448, PM519, PM520)

4.1.1 Regional Geology The Malay Basin is a **Cenozoic rift basin** in the southern South China Sea, ~500 km long × 200 km wide, trending NW-SE. It is Malaysia’s most prolific hydrocar-

bon province, contributing ~40% of national resources with cumulative discoveries exceeding 12 billion barrels of oil equivalent (USGS, 1999).

Tectonic Evolution: 1. **Late Eocene–Oligocene (Syn-rift):** Extensional half-grabens filled with lacustrine shales and continental clastics (Groups M, L, K) 2. **Early Miocene (Sag/Transition):** Fluvial-deltaic systems prograde into the basin (Groups J, I, H) 3. **Middle–Late Miocene (Inversion):** Transpressional inversion creates E-W trending anticlines — the primary structural traps (Groups G, F, E, D) 4. **Pliocene–Recent (Post-inversion):** Marine transgression, regional subsidence (Groups A, B, C)

Petroleum Systems: - **Source Rocks:** Oligocene–Miocene lacustrine shales (Groups L, M) and Miocene coaly shales (Groups H, I). Two source types: Type I/II lacustrine kerogen and Type III terrestrial/coaly kerogen. - **Reservoirs:** Fluvial-deltaic sandstones (Groups D, E, F, H, I) with porosities 15–25%. Deep Pre-Tertiary carbonates represent an emerging play. - **Seals:** Basin-wide marine shales (K shale is regionally extensive), intra-formational shales. - **Traps:** Inversion anticlines (E-W trending), fault-dependent closures, stratigraphic traps. - **Migration:** Vertical along deep-seated faults, lateral along carrier beds.

Key Insight: The basin is **mature for conventional plays**. Remaining Yet-to-Find (YTF) estimated at ~2–3 Bboe (Madon, 2021), primarily in: - Deep Pre-Tertiary carbonates (new play concept) - Stratigraphic traps in Groups F–I (requires advanced seismic) - Basin-flank plays on the NE ramp margin

4.1.2 Block PM440 — Near-Field Static

Parameter	Value
Area	4,840 km ²
Water Depth	50–60 m
PSC	EPT
MWC	Phase 1: 3 years, drill 1 well
Data	myPro 3D MC3D Repro (3,000 sq km)
MPM Prospectivity	Near-field, static play

Geological Narrative: PM440 is positioned in the **central Malay Basin** — a mature area with proven petroleum systems. The block sits within the main oil-prone fairway where Groups H and I source rocks are in the oil window. The near-field setting means existing infrastructure (platforms, pipelines) may be accessible for early monetization.

Play Types: - Conventional Miocene clastic inversion anticlines (proven) - Potential stratigraphic traps in deltaic systems - Possible Pre-Tertiary carbonate targets at depth

Risk Assessment: - **Source:** LOW — proven regional kitchen - **Reservoir:** MEDIUM — conventional targets well-understood, but quality varies - **Trap:** MEDIUM — need to identify undrilled closures - **Seal:** LOW — K shale regionally effective - **Charge:** LOW — active kitchen

BID RECOMMENDATION: CONDITIONAL BID — if 3D seismic data room confirms undrilled structural closures with >50 MMboe potential.

4.1.3 Block PM447 — New Play Ideas

Parameter	Value
Area	4,325 km ²
Water Depth	20–60 m
PSC	EPT
MWC	Phase 1: 3 years, drill 1 well + license MC3D Repro
Data	Mypro 3D MC3D Repro (3,000 sq km)
MPM Prospectivity	Portfolio being interpreted on recently acquired Penyu MC3D

Geological Narrative: PM447 is a **high-interest block** because MPM has identified it as hosting **new play ideas** beyond the conventional Miocene clastic fairway. The block is positioned where the **Penyu Basin** meets the Malay Basin, creating a unique geological junction. Recent MC3D seismic acquisition (Penyu MC3D) has been completed and is being interpreted — this suggests MPM sees material upside.

Play Types: - **Shallow Pre-Triassic carbonate play** — MPM specifically references this as a target - Conventional Miocene clastic inversion anticlines - Possible syn-rift clastic plays in deeper Groups K, L, M

Risk Assessment: - **Source:** LOW-MEDIUM — Penyu Basin source rocks less proven than central Malay Basin - **Reservoir:** MEDIUM — Pre-Tertiary carbonates are fracture-dependent (3–8% porosity base) - **Trap:** MEDIUM — carbonate buildups may be structurally complex - **Seal:** MEDIUM — need regional seal above carbonate section - **Charge:** MEDIUM — migration pathways from Malay Basin kitchen uncertain

BID RECOMMENDATION: BID — The Pre-Tertiary carbonate play represents a **material new play concept** in a mature basin. The MC3D Repro data reduces technical risk. If carbonates deliver, upside could be >100 MMboe per structure.

4.1.4 Block PM448 — Shallow Pre-Triassic

Parameter	Value
Area	—
Water Depth	—
PSC	EPT
MWC	Phase 1: 3 years, drill 1 well
Data	myPro 3D available
MPM Prospectivity	Shallow Pre-Triassic opportunity, drill-ready

Geological Narrative: PM448 is described as having a “**drill-ready Shallow Pre-Triassic opportunity**” — this is significant because it means MPM has already identified a specific prospect and the block is positioned for near-term drilling. The Pre-Tertiary carbonate play in the Malay

Basin is analogous to proven plays in the Penyu Basin and represents a step-out from conventional Miocene targets.

Play Types: - **Pre-Tertiary carbonate** — primary target, shallow enough for efficient drilling - Conventional Miocene clastic (secondary)

Risk Assessment: - **Source:** MEDIUM — Pre-Tertiary charge model needs validation - **Reservoir:** MEDIUM — carbonate porosity depends on fracture development and dissolution - **Trap:** LOW-MEDIUM — drill-ready suggests structural closure mapped on seismic - **Seal:** MEDIUM — Pre-Tertiary seal integrity needs verification - **Charge:** MEDIUM — migration from deeper kitchen possible

BID RECOMMENDATION: BID — Drill-ready status and low MWC (1 well) make this a high-value target. Even a moderate discovery (30–50 MMboe) would be commercial under EPT terms.

4.1.5 Block PM519 — Deep Pre-Tertiary

Parameter	Value
Area	5,629 km ²
Water Depth	5–75 m
PSC	EPT
MWC	Phase 1: 4 years, drill 1 well + 2D OBN
Data	myPro 3D negligible; Penyu MC3D 1,300 sq km
MPM Prospectivity	Pre-Tertiary opportunities relatively shallow in PM519 (1–2 km dbml)

Geological Narrative: PM519 hosts Pre-Tertiary targets that MPM notes are “**relatively shallow**” at 1–2 km below mud line — this is drillable with conventional jack-up rigs. However, the data availability is described as “negligible” for myPro 3D, with only Penyu MC3D (1,300 km²) covering a portion of the block.

Play Types: - Pre-Tertiary carbonate/clastic — primary target - Conventional Miocene clastic — secondary

Risk Assessment: - **Source:** MEDIUM-HIGH — Pre-Tertiary charge model speculative - **Reservoir:** HIGH — fracture porosity (3–8%) may not be sufficient - **Trap:** MEDIUM — structural mapping possible on available seismic - **Seal:** HIGH — Pre-Tertiary seal integrity uncertain - **Charge:** HIGH — migration pathways unclear

BID RECOMMENDATION: NO BID — Poor data coverage (“negligible” 3D), combined with high geological risk on Pre-Tertiary targets. The 4-year Phase 1 with 2D OBN acquisition adds cost. Better opportunities exist in PM447 and PM448.

4.1.6 Block PM520 — Deep Pre-Tertiary (>5 km)

Parameter	Value
Area	6,110 km ²
Water Depth	5–75 m
PSC	EPT
MWC	Phase 1: 4 years, drill 1 well + 500 sq km 3D TZ seismic
Data	Negligible
MPM Prospectivity	Pre-Tertiary targets very deep (>5 km dbml)

Geological Narrative: PM520 targets Pre-Tertiary plays at **>5 km below mud line** — this puts the targets in the **high-pressure, high-temperature (HPHT) regime** where drilling costs escalate dramatically and reservoir quality is severely compromised. The block also requires acquisition of 500 sq km of Transition Zone (TZ) 3D seismic — a significant upfront investment.

Play Types: - Pre-Tertiary carbonate/clastic at >5 km depth — extreme HPHT

Risk Assessment: - **Source:** HIGH — over-mature at these depths - **Reservoir:** VERY HIGH — extreme compaction and diagenesis at >5 km - **Trap:** HIGH — structural uncertainty at depth - **Seal:** HIGH — cap rock integrity under HPHT conditions - **Charge:** HIGH — timing and migration highly uncertain

BID RECOMMENDATION: STRONG NO BID — >5 km targets with negligible data and HPHT conditions represent **prohibitive technical and financial risk**. A single well could cost >\$50M with very low probability of success. This block requires technology that may not exist commercially.

4.2 WEST SARAWAK BASIN BLOCKS (SK330, SK331, SK404)

4.2.1 Regional Geology The West Sarawak Basin is an **emerging petroleum province** located offshore Sarawak, Malaysian Borneo. It forms part of the greater Sarawak Basin system that has produced hydrocarbons from the Baram Delta Province and Luconia Platform.

Tectonic Setting: - Complex **transtensional deformation** during the Tertiary related to India-Indochina collision - Sinistral strike-slip zones (Sabah Shear, West Baram-Tinjar Lines, Lupar Line) - NE-SW trending anticlines formed by compression and wrenching

Petroleum Systems: - **Source Rocks:** Coaly shales and organic-rich claystones of the Nyalau/Setap Formation (Oligocene–Miocene), TOC values 1.54–70 wt% - **Reservoirs:** Miocene deltaic and shallow marine sandstones (Miri/Belait/Seria Formations), Oligocene carbonates - **Traps:** NE-SW trending anticlines, fault-dependent closures, carbonate buildups - **Proven Plays:** Miocene clastic, Oligocene carbonate, Pre-Tertiary basement

Key Insight: West Sarawak is **less explored than the Malay Basin** but has proven hydrocarbon systems nearby. The challenge is **CO risk** in the Luconia area and the need for better understanding of source rock distribution and migration pathways.

4.2.2 Block SK331 — West Sarawak Emerging

Parameter	Value
Area	2,835 km ²
Water Depth	70–200 m
PSC	EPT (2 phases)
MWC	Phase 1: 3 years, drill 1 well + license Beicip MC PH2
Data	myPro 3D (tbc sq km)
MPM Prospectivity	Emerging West Sarawak

Geological Narrative: SK331 is positioned in the **emerging West Sarawak fairway** where recent exploration has shown encouraging results. The block benefits from Beicip multiclient study data (PH2), which provides regional geological context. Water depths of 70–200 m are manageable with modern jack-up or semi-submersible rigs.

Play Types: - Miocene clastic — deltaic and shallow marine sandstones - Oligocene carbonate — potential buildups - Possible stratigraphic traps

Risk Assessment: - **Source:** MEDIUM — coaly source rocks regionally present but distribution uncertain - **Reservoir:** MEDIUM — Miocene sandstones proven in nearby fields - **Trap:** MEDIUM — anticlinal structures mapped on seismic - **Seal:** MEDIUM — Miocene shales provide regional seal - **Charge:** MEDIUM-HIGH — migration pathways from kitchen areas need validation

BID RECOMMENDATION: BID WITH PARTNER — reasonable risk-reward if a partner with Sarawak exploration experience is secured. The Beicip study data reduces geological uncertainty.

4.2.3 Block SK330 — Deepwater West Sarawak

Parameter	Value
Area	5,038 km ²
Water Depth	120–300 m
PSC	EPT (2 phases)
MWC	Phase 1: 3 years, drill 1 well + license Beicip MC PH2 + license 3,875 sq km TGS MC3D (\$5M USD license fee)
Data	TGS MC3D 3,875 sq km

Geological Narrative: SK330 has the **best seismic data coverage** of all MBR2026 blocks — 3,875 km² of TGS multiclient 3D seismic. However, the **\$5M USD data license fee** is a significant upfront cost. Water depths of 120–300 m require semi-submersible rigs or drillships, escalating well costs to \$40–60M per well.

Play Types: - Miocene clastic — deepwater fan and turbidite systems - Oligocene carbonate — potential reef buildups - Stratigraphic traps in deepwater setting

Risk Assessment: - **Source:** MEDIUM — same regional source kitchen - **Reservoir:** MEDIUM-HIGH — deepwater reservoir quality variable - **Trap:** MEDIUM — structural and stratigraphic traps possible - **Seal:** LOW-MEDIUM — deepwater shales effective - **Charge:** HIGH — long migration distances possible

BID RECOMMENDATION: CONSIDER WITH CAUTION — excellent data but high entry cost (\$5M license + deepwater well). Only viable for a well-capitalized operator with deep-water capability. Recommend farm-in strategy rather than direct bid.

4.2.4 Block SK404 — West Sarawak Primary

Parameter	Value
Area	4,829 km ²
Water Depth	40–100 m
PSC	EPT (2 phases)
MWC	Phase 1: 3 years, drill 1 well + license Beicip MC PH1
Data	Mypro 3D (tbc) + Beicip MC PH1

Geological Narrative: SK404 is the **most accessible West Sarawak block** — shallow water (40–100 m) allows jack-up rig operations at significantly lower cost than SK330. The Beicip multicient PH1 study provides regional geological context. The shallow water setting also means proximity to existing Sarawak infrastructure.

Play Types: - Miocene clastic — deltaic and shallow marine - Oligocene carbonate — platform buildups - Potential near-field tie-back opportunities to existing Sarawak production

Risk Assessment: - **Source:** MEDIUM — regional source proven but local charge model needs work - **Reservoir:** MEDIUM — Miocene sandstones proven regionally - **Trap:** MEDIUM — structural closures expected - **Seal:** LOW-MEDIUM — regional shales effective - **Charge:** MEDIUM — shorter migration paths than SK330

BID RECOMMENDATION: BID — Best risk-reward ratio among West Sarawak blocks. Shallow water = lower drilling costs. Beicip study data available. Proximity to existing infrastructure. This is the **preferred entry point for West Sarawak exploration**.

4.3 SANDAKAN BASIN (SB304)

4.3.1 Regional Geology The Sandakan Basin is a **frontier basin** located offshore northeast Sabah. It is part of the greater Tarakan Basin system with proven hydrocarbons in nearby Indonesian waters (Badik, Aster, Tulip Fields — ~500 MMboe cumulative).

Tectonic Setting: - Located at the junction of the Sulu Sea and Celebes Sea - Complex tectonic history involving subduction, accretion, and strike-slip deformation - Miocene to Pliocene sedimentary fill overlying older basement

Petroleum Systems: - **Source Rocks:** Miocene and intra-Pliocene clastic source rocks (coaly and marine) - **Reservoirs:** Miocene to Eocene clastic, Oligocene carbonate, Pre-Tertiary basement

- **Traps:** Structural (anticlines, fault blocks) and stratigraphic - **Proven Plays:** Miocene clastic, Oligocene carbonate, Pre-Tertiary

Key Insight: The Sandakan Basin is **truly frontier** — limited well control, no 3D seismic in SB304, and geological uncertainty is high. The positive: proven petroleum system in the greater Tarakan Basin, and MPM has extended the bid deadline to June 2027, suggesting they expect this block to attract patient capital.

4.3.2 Block SB304 — Frontier Sandakan

Parameter	Value
Area	7,030 km ²
Water Depth	0–20 m (onshore to nearshore)
PSC	EPT (2 phases)
MWC	Phase 1: 3 years, acquire 500 sq km 3D OBN seismic + 1000 line-km 2D OBN
Data	No 3D seismic available
Bid Deadline	30 June 2027 (extended)

Geological Narrative: SB304 is the **highest-risk, highest-reward block** in MBR2026. It covers 7,030 km² in the Sandakan Basin — a frontier area with virtually no modern seismic data. The nearshore/shallow water setting (0–20 m) is favorable for operations but the lack of subsurface data means **the first operator must invest heavily in seismic acquisition before any drilling decisions can be made.**

Play Types: - Miocene clastic — inferred from regional geology - Oligocene carbonate — potential platform buildups - Pre-Tertiary basement — untested - Possible stratigraphic traps in deltaic systems

Risk Assessment: - **Source:** HIGH — unproven in this specific block - **Reservoir:** HIGH — no well control, reservoir quality unknown - **Trap:** HIGH — structural mapping impossible without 3D seismic - **Seal:** HIGH — seal presence and integrity unknown - **Charge:** VERY HIGH — petroleum system entirely speculative

BID RECOMMENDATION: STRONG NO BID — The combination of no 3D seismic, frontier basin status, high upfront seismic acquisition costs (\$15–25M for 3D OBN), and extreme geological uncertainty makes this block **uninvestable** without a farm-in partner who has proprietary data. The extended deadline (June 2027) confirms MPM expects limited interest.

5. DISCOVERED RESOURCE OPPORTUNITIES — ASSESSMENT

5.1 Cempaka Cluster (Peninsular Malaysia)

Parameter	Value
Fields	Saujana, Serok, Semangkok NAG, Semangkok Timur

Parameter	Value
Fiscal Term	SFA PSC
Location	Peninsular Malaysia, mature Malay Basin

Assessment: The Cempaka Cluster consists of **four discovered gas/condensate fields** in the mature Malay Basin. These are small fields that were previously sub-commercial under legacy fiscal terms but may be viable under the simplified SFA PSC. The fields are located in a proven petroleum system with existing infrastructure nearby.

Upside: Low exploration risk (discovered resources), near-field development potential, infrastructure access. **Downside:** Small field sizes may limit economics, gas pricing uncertainty.

RECOMMENDATION: BID — Low-risk, ready-to-develop opportunity. SFA terms designed for exactly this type of asset.

5.2 Teratai Cluster (Peninsular Malaysia)

Parameter	Value
Fields	Sotong, Malong, Anding, Aji, Aji Sentang, Parang, Naga Kecil, Naga Dalam
Fiscal Term	SFA PSC
Location	Peninsular Malaysia, Malay Basin

Assessment: The Teratai Cluster contains **eight fields** — the largest DRO by field count. This suggests either multiple small accumulations or a complex of related fields. The high field count may indicate aggregation economics — individually sub-commercial but collectively viable.

Upside: Multiple fields provide portfolio diversification, potential for phased development. **Downside:** Higher operational complexity, multiple well programs needed.

RECOMMENDATION: BID WITH PARTNER — Operational complexity requires a capable operator. Consider joint venture with a Malaysian service company.

5.3 Enggang Cluster (Sarawak)

Parameter	Value
Fields	Salam, Benum, Patawali, Gagau
Fiscal Term	SFA PSC
Location	Sarawak, formerly WL4-00 PSC

Assessment: The Enggang Cluster represents **previously producing fields** that were formerly under the WL4-00 PSC. These fields have production history, which is invaluable for reserves estimation and development planning. The fields are being re-offered, suggesting they have remaining upside that the previous operator did not capture.

Upside: Production history, known reservoirs, existing well data, redevelopment rather than green-field. **Downside:** Possible depletion, legacy infrastructure condition unknown, Sarawak regulatory complexity.

RECOMMENDATION: BID — Production history dramatically reduces subsurface risk. SFA terms make small field redevelopment viable.

5.4 NURI Funnel (Sarawak)

Parameter	Value
Fields	Nuang, Rompin, KT1, Endau, Mahkota E2, Serunai
Fiscal Term	SFA PSC
Location	Sarawak

Assessment: The NURI Funnel is a **six-field cluster** in Sarawak. The “Funnel” naming convention in MBR suggests a group of fields that feed into a common development concept (shared infrastructure, common processing).

Upside: Cluster economics, shared infrastructure reduces per-field cost. **Downside:** Requires significant upfront infrastructure investment.

RECOMMENDATION: CONDITIONAL BID — Depends on field sizes and infrastructure requirements. Need data room assessment.

5.5 Nilam Cluster (Sabah)

Parameter	Value
Fields	Trusmadi, Madalon, Papar, Padas, Grait
Fiscal Term	SFA PSC
Location	Sabah

Assessment: Five fields in offshore Sabah. Limited public information on these fields. Sabah operations typically involve higher logistics costs and regulatory complexity compared to Peninsular Malaysia.

RECOMMENDATION: CONDITIONAL BID — Requires detailed data room assessment. Sabah operations carry higher cost base.

5.6 Zamrud Funnel (Sabah)

Parameter	Value
Fields	Tembungo, SW Emerald, Danum, Rusa Timur, Sipadan
Fiscal Term	SFA PSC
Location	~100 km offshore Sabah, water depth 70–125 m

Assessment: The Zamrud Funnel is the **first DRO offered** in MBR2026 (launched January 2026 ahead of the main round). It includes the **Tembungo field** (previously operated by Petronas Carigali) and **Danum** (formerly part of Block SB412, awarded to PTTEP in 2018). The cluster is located ~100 km offshore Sabah in 70–125 m water depth.

Upside: Known producing history (Tembungo), proximity to PTTEP’s SB412 operations, multiple fields provide portfolio effect. **Downside:** Remote location, higher logistics costs, moderate water depths.

RECOMMENDATION: BID — Tembungo production history reduces risk. PTTEP’s nearby operations provide infrastructure optionality.

6. ECONOMIC SCREENING

6.1 Exploration Well Cost Estimates

Basin	Water Depth	Rig Type	Estimated Well Cost
Malay Basin (shallow)	20–75 m	Jack-up	\$15–25M
West Sarawak (mid)	40–200 m	Jack-up/Semi-sub	\$25–45M
West Sarawak (deep)	120–300 m	Semi-sub/Drillship	\$40–60M
Sandakan (nearshore)	0–20 m	Jack-up/Land rig	\$10–20M

6.2 Minimum Work Commitment Cost Estimates

Block	Phase 1 MWC	Estimated Cost (USD)
PM440	1 well	\$15–25M
PM447	1 well + MC3D Repro license	\$20–30M
PM448	1 well	\$15–25M
PM519	1 well + 2D OBN	\$25–35M
PM520	1 well + 500 km ² 3D TZ	\$35–50M
SK330	1 well + Beicip + TGS MC3D (\$5M)	\$50–70M
SK331	1 well + Beicip	\$30–50M
SK404	1 well + Beicip	\$30–45M
SB304	500 km ² 3D OBN + 1000 km 2D	\$20–30M (seismic only)

6.3 EMV Screening (P50, 10% discount rate)

For a typical Malay Basin near-field discovery:

Parameter	Conservative	Base	Optimistic
STOIIP (MMstb)	50	150	400
Recovery Factor	20%	30%	40%
Recoverable (MMstb)	10	45	160
Oil Price (\$/bbl)	\$65	\$80	95 <i>DevelopmentCapex</i> (M)
EMV (riskd, POS 0.35)	-\$18	\$63	\$228

Key Insight: Near-field Malay Basin opportunities (PM447, PM448) have **positive EMV** at base case, making them economically viable under EPT terms.

7. BID/NO-BID RECOMMENDATIONS

7.1 Priority Tier — RECOMMENDED TO BID

Block	Type	Basin	Rationale
PM447	Exploration	Malay	New Pre-Triassic carbonate play, MC3D data, near-field, moderate MWC
PM448	Exploration	Malay	Drill-ready Pre-Triassic target, low MWC, proven petroleum system
Cempaka Cluster	DRO	Malay	Ready-to-develop, SFA terms, low exploration risk
Enggang Cluster	DRO	Sarawak	Production history, redevelopment opportunity, SFA terms

7.2 Second Tier — CONSIDER WITH PARTNERS

Block	Type	Basin	Rationale
SK404	Exploration	W. Sarawak	Best entry point for West Sarawak, shallow water, Beicip data
SK331	Exploration	W. Sarawak	Emerging play, 3D data available, partner required
Zamrud Funnel	DRO	Sabah	Tembungo production history, multiple fields

Block	Type	Basin	Rationale
Teratai Cluster	DRO	Malay	Multiple fields, complex operations, partner needed

7.3 Third Tier — NOT RECOMMENDED

Block	Type	Basin	Rationale
SB304	Exploration	Sandakan	Frontier, no 3D, extreme risk, high seismic cost
PM519	Exploration	Malay	Negligible data, Pre-Tertiary at uncertain depth
PM520	Exploration	Malay	>5 km targets, HPHT, prohibitive risk
PM440	Exploration	Malay	Static play, limited upside vs PM447/PM448
SK330	Exploration	W. Sarawak	\$5M data license + deepwater well = too expensive
NURI Funnel	DRO	Sarawak	Insufficient public data, conditional
Nilam Cluster	DRO	Sabah	Insufficient public data, conditional

8. RISK REGISTER

8.1 Key Risks

#	Risk	Probability	Impact	Mitigation
1	Pre-Tertiary carbonate play failure	MEDIUM	HIGH	Multiple targets across PM447/PM448; don't commit to both unless data confirms
2	Gas price collapse	LOW	HIGH	Focus on oil-prone targets; DROs with oil production history preferred

#	Risk	Probability	Impact	Mitigation
3	Partner default	LOW	MEDIUM	Due diligence on partner financials; escrow arrangements
4	Regulatory delay	MEDIUM	MEDIUM	Engage Malaysian counsel early; maintain MPM relationship
5	Seismic data quality	MEDIUM	HIGH	License premium multiclient data; budget for reprocessing
6	Drilling cost overrun	MEDIUM	MEDIUM	Fix-price drilling contracts; jack-up rig commitment
7	CO risk (Sarawak)	HIGH	HIGH	Geochemical screening before drilling; avoid Luconia high-CO fairway

8.2 Portfolio Risk Matrix

	LOW RISK	MEDIUM RISK	HIGH RISK
HIGH REWARD	Cempaka DRO	PM447, PM448	SB304
MEDIUM REWARD	Enggang DRO	SK404, SK331	SK330
LOW REWARD	Zamrud DRO	Teratai Cluster	PM520

9. APPENDICES

Appendix A: MBR 2026 Timeline

Milestone	Date
MBR 2026 Launch	10 February 2026
Data Room Open	10 February 2026
DRO Data Room Open	February 2026
Bid Submission Deadline (all except SB304)	1 October 2026
Bid Submission Deadline (SB304)	30 June 2027
PSC Award Target	February 2027
PSC Signing Target	April 2027

Appendix B: Contact Information

Query Type	Contact
MBR General	mbr-query@petronas.com.my
myPROdata Subscription	https://myprodata.petronas.com/register
MBR Portal	https://www.malaysiabidround.com
Data Room	Accessible via myPROdata subscription

Appendix C: Petroleum Systems Summary

Basin	Source	Reservoir	Seal	Trap	Charge Risk
Malay	Lacustrine shales (L,M) + coaly shales (H,I)	Fluvial-deltaic sandstones (D,E,F,H,I)	K shale + intra-formational	Inversion anticlines, fault closures	LOW
W. Sarawak	Coaly shales (Nyalau/Setap)	Miocene deltaic sands, Oligocene carb	Miocene shales	NE-SW anticlines, carbonate buildups	MEDIUM
Sandakan	Inferred Miocene/Pliocene clastics	Miocene elastics, Oligocene carb	Regional shales	Structural + stratigraphic	HIGH

Appendix D: Economic Assumptions

Parameter	Value	Source
Oil Price (Brent)	\$80/bbl (base)	Market consensus
Gas Price	\$8/MMBtu	Malaysian hub price
Discount Rate	10%	Standard exploration
Exploration Well (Jack-up)	\$15–25M	Industry benchmarks
Exploration Well (Semi-sub)	\$40–60M	Industry benchmarks
Development Capex (per well)	\$15–25M	Malay Basin analogues
Operating Cost	\$3–5/boe	Malaysian shallow water
POS (exploration well)	0.30–0.40	Basin maturity dependent

Appendix E: Evidence Quality Legend

Label	Meaning	Confidence Cap
OBS	Observed — direct measurement or data room inspection	0.90
DER	Derived — computed from data (e.g., volumetrics)	0.85
INT	Interpreted — geological synthesis from multiple sources	0.75
SPEC	Speculated — projection based on analogue or model	0.60

DOCUMENT CONTROL

Version	Date	Author	Changes
1.0	2026-07-09	FORGE (000Ω)	Initial draft — full MBR2026 portfolio assessment

Evidence Sources: - PETRONAS MPM official launch statement (11 Feb 2026) - PETRONAS myPROdata MBR portal (accessed 9 Jul 2026) - MBR 2026 Summary Draft (Scribd, Apr 2026) - USGS OF99-50: Petroleum Systems of the Malay Basin Province - Madon (2021): Five Decades of Petroleum Exploration in the Malay Basin - Multiple industry sources (Reuters, Wood Mackenzie, Welligence, New Vision) - GEOX Basin Profile: Malay Basin (v2026.06.22)

Epistemic Labels: All geological claims labeled OBS/DER/INT/SPEC throughout. **Confidence Cap:** 0.90 for OBS, 0.85 for DER, 0.75 for INT, 0.60 for SPEC.

Prepared by FORGE (000Ω) — arifOS Federation DITEMPA BUKAN DIBERI — Forged, Not Given For: Muhammad Arif bin Fazil (F13 Sovereign, 888)